# Fabien Depoilly

## Personal Information


**Name:** Depoilly
**First name:** Fabien
**Date of birth:** March 23, 1999 (27 years old)
**Nationality:** French
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## Professional Experience

**Polytech Lyon** *December 2025 – August 2026*
*Temporary Teaching and Research Assistant (ATER)*

**Lyon 1 University Claude Bernard** *December 2022 – December 2025*
*PhD Candidate, with Supplementary Teaching Duties (ACE)*

## Academic Background

**Lyon 1 University Claude Bernard** *December 2022 – June 2026*
*PhD Thesis*

- **Defended on June 30, 2026**
- **Title:** Stability of generalised Newtonian fluid film flows down an inclined plane.
- **Advisor:** Hamda Ben Hadid (Professor)
- **Co-supervisors:** Séverine Millet (Associate Professor), Simon Dagois-Bohy (Associate Professor)
- **Research Unit:** Laboratory of Fluid Mechanics and Acoustics (LMFA, UMR CNRS 5509)

**Lyon 1 University Claude Bernard** *September 2020 – September 2022*
*Master's degree in Mechanics, Fluid Mechanics and Energetics track*

**Le Mans Université** *September 2017 – June 2020*
*Bachelor's degree in Engineering Sciences, Acoustics track*

## Research Activity

My research focuses on the stability of thin-film flows of non-Newtonian fluids down an inclined plane, relevant to numerous natural and industrial contexts (avalanches, mudflows, coating processes, etc.), and prone to becoming unstable. Given the diversity of rheological behaviours, my work aims to establish a unifying theoretical framework — independent of any specific rheological law — confronted with experimental data obtained on an inclined channel.

**From left to right:** surface waves on Carbopol, a surface coating process (Kistler & Schweizer (1997)), surface waves on melted chocolate.

## International Conference Participation

**11$^{th}$ International Bifurcation and Instabilities in Fluid Dynamics Symposium (BIFD)** *July 6-9, 2026*
*L'Aquila, Italy*

- **Presentation type:** Oral presentation
- **Title:** Short-wave Linear Stability of Non-Newtonian Fluids Down a Slope
- **Co-authors:** Fabien Depoilly, Séverine Millet, Simon Dagois-Bohy, Hamda Ben Hadid

**GDR TransInter Workshop** *June 10-12, 2025*
*Aussois, France*

- **Presentation type:** Oral presentation
- **Title:** Stability of non newtonian fluids down a slope
- **Co-authors:** Fabien Depoilly, Simon Dagois-Bohy, Séverine Millet, François Rousset, Hamda Ben Hadid

**10$^{th}$ International Bifurcation and Instabilities in Fluid Dynamics Symposium (BIFD)** *June 24-28, 2024*
*Edinburgh, Scotland*

- **Presentation type:** Oral presentation
- **Title:** Primary instability of roll waves on thin films of non-Newtonian fluids down a slope.
- **Co-authors:** Fabien Depoilly, Simon Dagois-Bohy, Séverine Millet, François Rousset, Hamda Ben Hadid

**Fluids and Complexity III** *December 6-8, 2023*
*Nice, France*

- **Presentation type:** Poster presentation
- **Title:** Linear stability of a generalised Newtonian fluid down an incline: unifying the roll waves
- **Co-authors:** Fabien Depoilly, François Rousset, Séverine Millet, Hamda Ben Hadid, Simon Dagois-Bohy

## Teaching Activities

- **Lyon 1 University Claude Bernard, Department of Mechanics**
  - 2025: Introduction to Fluid Mechanics, 2nd-year Bachelor's in Mechanics (Tutorials)
  - 2023 – 2025: Scientific Computing, 3rd-year Bachelor's in Mechanics (Labs)
  - 2023 – 2025: Finite Difference Methods for PDEs in Mechanics, 1st-year Master's in Mechanics (Tutorials/Labs)
- **Polytech Lyon**
  - 2025: Fluid Mechanics 2, 3rd-year Mechanical Engineering (Tutorials/Labs)
  - 2025: Mathematical Tools for Engineers 2, 3rd-year Mechanical Engineering (Tutorials)
  - 2023 – 2024: Basic Numerical Methods, 3rd-year Mechanical Engineering (Labs)
  - 2023: Numerical Methods for Mechanics 1, 3rd-year Mechanical Engineering (Labs)

## Responsibilities

**Elected member of the Research Commission, Academic Council** *2022 - 2024*
*Lyon 1 University Claude Bernard*

## Publications

[1] Depoilly, F., Millet, S., Hadid, H. B., Dagois-Bohy, S., Rousset, F., "Unifying the roll waves," *PLOS ONE*, vol. 19, no. 11, Nov. 19, 2024, Publisher: Public Library of Science, ISSN: 1932-6203. DOI: 10.1371/journal.pone.0310805